

QA Final Test Summary Report

Project Name: HTU QA Capstone Project (SauceDemo & DummyJSON)

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1. Executive Summary

This report summarizes the comprehensive testing activities performed on the **Swag Labs (SauceDemo)** e-commerce website and **DummyJSON API**.

The testing approach followed the strategy defined in the **Test Plan**, covering Manual, Automation, and Performance testing.

Overall Decision:

- **API (Backend): PASSED** (Stable & Performant).
 - **UI (Frontend): FAILED** (Critical Defects Found in problem_user flow).
 - **Recommendation:** Fix the critical Checkout blocker before release.
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2. Test Scope & Objectives

As defined in the Test Plan, the testing focused on the **End-to-End (E2E)** flow:

- **In-Scope:** Login Page, Inventory Page, Cart Page, Checkout Process, and Sidebar Functionality.
 - **Test Data:** Used standard_user for happy paths and problem_user for defect discovery.
 - **Out-of-Scope:** Payment Gateway & Backend Database Verification.
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3. Test Execution Results

3.1 Manual Testing (Functional)

- **Total Test Cases:** 20+ Scenarios.
- **Key Scenarios Verified:**
 - **SD01:** Verify sign-in functionality with accepted credentials (Passed).
 - **SD03:** Verify sign-in with blank username (Passed).
 - **SD04:** Verify Remove button functionality (Passed).

3.2 Defect Analysis (Bugs Found)

Critical functional defects were identified during the testing of the problem_user persona.

- **Total Bugs Logged:** 8
- **Top Critical/High Defects:**
 1. **Critical:** *Checkout blocked (Last Name field disabled)*. The user cannot enter the last name, preventing the checkout process from completing.
 2. **High:** *Inability to add all items to cart*. Buttons for adding items do not function correctly for specific products.

3.3 API Performance Testing (k6)

- **Target:** DummyJSON API
- **Load:** 15 Virtual Users for 30 seconds.
- **Results:**
 - **Error Rate:** 0.00%
 - **Response Time (p95):** ~323ms
 - **Conclusion:** The backend handles concurrent load efficiently without errors.

4. Automation Coverage

- **API Automation:** Postman collection created for CRUD operations (GET, POST, PUT, DELETE).
 - **UI Automation:** Selenium/Cypress scripts implemented to cover the critical Login -> Checkout path.
 - **CI/CD:** GitHub Actions pipeline configured to run regression tests on every push.
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5. Conclusion

The testing cycle confirms that while the **Backend (API)** is robust and ready, the **Frontend (UI)** requires immediate attention. The **Critical Bug (Checkout Blocked)** is a release blocker.

Sign-off:

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